

The intergenerational mobility effects of higher education expansion in China

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ABSTRACT

This study examines the possible effects of China's college expansion policy on intergenerational income mobility using data from the China Family Panel Studies (CFPS) from 2010 to 2018. The results show that, first, higher education expansion policy does not significantly improve intergenerational income mobility. Instead, higher education expansion policy may increase the intergenerational income correlations in cities, and areas with abundant resources for higher education opportunities. Meanwhile, higher education expansion policy may trigger the transmission of intergenerational poverty in rural areas, economically underdeveloped areas, and areas with less abundant resources or opportunities for higher education. Second, both elite higher education and postgraduate education significantly improve individuals' income rank and intergenerational income mobility, but family background still plays a strong role in influencing the income rank of children. Third, there is a Gatsby curve in China in which an increase in income inequality is associated with a decline in intergenerational income mobility. This paper argues for the importance of expanding higher education while highlighting the limitations of this policy.

1. Introduction

Corak (2013) proposed the 'Gatsby curve' to show that countries with high inequality generally have lower intergenerational mobility and that greater income disparities among residents can inhibit intergenerational mobility. As an important mechanism for escaping poverty and breaking down class solidification, education, especially higher education, plays a positive role in enhancing individual mobility, social progress, and intergenerational mobility. The higher individuals' relative level of education is, the more competitive they are in the labour market. Thus, as a key to social stratification, education contributes to social advancement through access to quality occupations and higher income (Collins, 1977).

China implemented the college expansion policy in 1999. The

number of students enrolled in higher education institutions increased from 1,596,800 in 1999 to 10,013,200 in 2021, and the gross enrolment rate in higher education increased from 10.5 % to 57.8 %. The transformation of elite university systems into systems of mass higher education was successfully realized. However, the additional higher education enrolment opportunities were driven by enrolment increases at local ordinary colleges and universities, and the increase in enrolment opportunities at top or key colleges and universities was not significant.¹ The entry of a large number of graduates from ordinary colleges and universities into the labour market appeared to weaken the signal value of higher education qualifications and has led to social problems such as graduate unemployment, devaluation of qualifications and class entrenchment. The data show that the intergenerational elasticity index and the intergenerational rank index of different generations in China

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¹ Top or key universities refer to the "Project 985" and "Project 211" in China. "Project 985" is a project aimed at founding world-class universities proposed by Comrade Jiang Zemin in his speech at the 100th anniversary celebration of Peking University in May 1998, and 39 universities have been identified, including Peking University, Tsinghua University. "Project 211" was officially approved by the State Council in November 1995, with the aim of building 100 universities and some major fields of study for the 21st century.

Table 1
Intergenerational income mobility across birth generations.

Intergenerational income mobility index	Cohorts			
	50s-generation	60s-generation	70s-generation	80s-generation
age	0.42	0.38	0.32	0.38
rank	0.46	0.47	0.45	0.43

have not changed significantly (see Table 1). Furthermore, the intergenerational income mobility of the post-1980s group,² which has been significantly affected by the expansion of higher education policies, has not improved significantly compared with previous generations, indicating that the implementation of the higher education expansion policy may not have significantly improved intergenerational income mobility in China.³

Prior to the college expansion policy, the relationship between higher education and intergenerational mobility was strong. People who received higher education largely were allocated to or found better or higher-paid jobs than their parents and thus achieved upward mobility. During this period, higher education played a major role in promoting intergenerational mobility, especially for the lower classes. Many children from low-income families discovered that a college degree allowed them to break free from the limitations of their family background and move forwards. However, following the expansion of higher education in 1999, the main function of higher education has begun to shift from producing a small number of social elites to producing talent adapted to the socioeconomic and industrial development of society.

At the beginning of the higher education expansion policy, which focused on expanding the scale of development of higher education at the expense of quality improvement, the personnel training model in higher education was relatively closed. It emphasized theoretical training rather than practice, and a variety of triggers were superimposed on the weakening of the higher education signalling function of the diploma. In addition, the speed of China's economic and industrial structure transformation lags behind the speed of higher education expansion. For a long period, China's industrial development has been based mainly on the manufacturing industry, particularly the low-end manufacturing industry. The high-end manufacturing and productive service industry started late and has lagged behind, and higher human capital is associated mainly with high-end manufacturing or high value-added industries. Due to the limited supply of jobs in the main labour market, when the supply of higher human capital is much greater than the demand for jobs, this will lead to a decline in the relative rate of return to higher education (Ma et al., 2017), the depreciation of higher education diplomas, and a significant increase in the difficulty of university graduates obtaining quality occupations and labour

remuneration in line with equivalent qualifications before the expansion of higher education. To recruit the highest-quality graduates, employers even "raise the bar" by artificially increasing the qualification threshold for job applications and even explicitly requiring the recruitment of only graduates from "double first-class" or elite universities and overseas students. This phenomenon has exacerbated the mismatch and over-education in the labour market and raised widespread concern about the devaluation of university degrees.

China's higher education expansion policy was initially driven by the governments and was implemented in a relatively "market-based" manner. Therefore, the intergenerational mobility effects of higher education expansion may differ across regions. For less economically developed regions or regions with disadvantaged higher education resources, access to higher education is limited by resource constraints, which indirectly affects intergenerational mobility opportunities. It remains to be empirically tested whether higher education expansion policies promote intergenerational mobility through educational equity, how the functional effects of higher education on intergenerational income change across regions, and whether higher education weakens intergenerational transmission effects and income inequality.

To answer these questions, instead of following the traditional approach of other scholars and measuring intergenerational income by directly limiting the age of the sample and using persistent income, this study considered potential sample selection bias and used the Heckman sample selection model to address cohabitation bias and temporary migration bias. Further empirical tests were conducted to examine the direct and heterogeneous effects of higher education expansion on intergenerational income mobility. This study found that, first, the expansion of higher education increased intergenerational income correlation and reduced intergenerational income mobility. Specifically, the phenomenon of intergenerational income persistence was strengthened in urban areas, eastern areas, and areas with abundant higher education resources; at the same time, intergenerational income mobility in rural areas, less developed areas in the central and western regions, and areas with abundant higher education resources increased. Second, this paper empirically investigates the internal logic of the influence of higher education on intergenerational income correlations and finds that elite higher education and postgraduate education effectively improve individual income rankings and mitigate the impact of family capital on children's income. However, there is no evidence of weakening intergenerational income persistence. Finally, we verify that there is a Gatsby curve in China and that higher income inequality is accompanied by intergenerational inequality.

The main contents of the article are organized as follows. The first part presents the introduction, and the second part summarizes the research progress of relevant scholars. The third part describes the research design, which includes the treatment of sample selection bias and model setting. The fourth part is the empirical analysis, which examines the effects of higher education expansion on intergenerational mobility and heterogeneity effects, further discusses how elite higher education and postgraduate education affect the income rank of children and intergenerational correlations, and finally verifies the existence of the Gatsby curve in China. The fifth section provides conclusions and policy implications.

2. Literature review

Different schools of thought have different views on the relationship between education and intergenerational mobility. Durkheim (1956), the founder of functionalist thought, is the sociologist who has paid the most attention to the issue of education. He suggested that education has the function of socialization and can help individuals form the correct social values or concepts to develop an independent social personality and thus achieve social integration and functional control. The value of a degree is related to rarity and relative educational attainment, with higher degrees signalling a place in a higher class and functioning as a

² The legal age for enrolling in primary school in China is 7 years old, the age for enrolling in higher education is 19 years old, and the year in which the higher education expansion policy was implemented was 1999, meaning that all groups born after 1980 were affected by the higher education expansion policy.

³ The immediate reason for introduce higher education expansion policy in 1999 was to avoid the impact of the Asian financial crisis, which affected China's economic growth in 1997 and 1998. Especially after March 1999, the negative impact of the Asian financial crisis on China fully appeared. Coupled with the reform of state-owned enterprises brought a large number of laid-off employees, more than 3 million high school graduates every year will also form a new employment pressure. Therefore, the government decided to implement a proactive fiscal policy to stimulate domestic demand and ease the employment pressure, and the higher education expansion is a matter of common expectation. The implementation of higher education expansion, while facilitating the rapid development of higher education, created a mismatch between the supply of higher education and the effective demand in the labor market, which to some extent exacerbated the inflation of academic qualifications and the raising of academic barriers in the labor market.

key mediator to increase social mobility. Marginson (2018) argued that the scope of higher education in creating equality of opportunity cannot be overstated and that family background, government resources and regional levels of economic development may all be key forces that affect social mobility. Collins (1977) agreed that a diploma is a necessary condition for high income and access to higher classes, but he further noted that people compete fiercely for diplomas to obtain better jobs; thus, schools are important sites of competition for upper-class interests and tools for both status and cultural reproduction. Marginson (2016) noted that societies with more equal income distributions and greater social mobility have higher rates of participation in higher education and that in the US, the children of families in which the father has higher education are 6.8 times more likely to attend higher education than children from families with no higher education.

As higher education continues to expand, the impact of educational expansion on intergenerational mobility has been widely debated. Some scholars argue that educational expansion can bring about equality in educational opportunities (Breen and Jonsson 2007; Chesters and Watson, 2013; Breen et al., 2009), reducing the correlation between family background and educational opportunity. However, some scholars have argued that educational expansion does not lead to equality of opportunity (Iannelli, 2011; Alon, 2009). The most prominent of these arguments are maximally maintained inequality (MMI) and effectively maintained inequality (EMI). The MMI hypothesis argues that educational expansion does not equalize the distribution of opportunity and that inequality in educational opportunity will persist as long as it is possible for the dominant class to improve its educational opportunities (Raftery and Hout, 1993). This hypothesis has been supported by many scholars (Alon, 2009; Shavit and Westerbeek 1998; Iannelli and Smyth, 2008). In contrast, the EMI hypothesis suggests that even when advantaged classes have access to higher education, inequalities in higher education opportunities will remain, and the pursuit of higher education by advantaged classes will shift from quantitative to qualitative inequalities due to the search for higher-quality higher education (Lucas, 2001). This hypothesis is supported by numerous empirical studies (Ayalon and Yogev, 2005; Yeung, 2013; Guo et al., 2019). As noted by Guo et al. (2019), after the expansion of higher education, educational opportunities, especially quality educational opportunities, were more likely to be taken by children from households in the resource-rich class, whereas the effect on households in the resource-poor class may be very limited. In the UK, the expansion of higher education has increased the gap in higher education attainment between rich and poor children and reduced intergenerational educational mobility (Blanden and Machin, 2013).

The intergenerational elasticity of earnings or income is a common measure of intergenerational mobility (Dahl and DeLeire, 2008; Chetty et al., 2014). Scholars have used characteristic variables such as parental education, parental race, and occupation as instrumental variables to correct for estimation bias (Solon, 1992; Lefgren et al., 2012) because measuring intergenerational mobility in terms of income is problematic due to life-cycle and transitory income shocks. Depending which type of data or measure is used, different conclusions are reached. For example, some scholars argue that China has a low level of intergenerational mobility with replication and entrenchment between intergenerational classes and a pronounced trend of widening income inequality and income polarization (Yuan, 2017;). Others argue that China's economic development and deepening reforms have been accompanied by a continuous increase in intergenerational mobility (Lv and Li, 2017).

Most of the literature focuses on the impact of China's higher education expansion policy on intergenerational mobility and examines the changes in access to higher education triggered by higher education expansion and the moderating effect of family capital on access to higher education. However, few studies have investigated the impact of higher education expansion on intergenerational income mobility. Intergenerational income mobility not only reflects the dynamic pattern of income distribution and the degree of equality of opportunity but is also an

important indicator of social fairness and justice and is more in line with the reality of the future development of China and the world. In addition, no research has explored how changes in the internal and external relationships of higher education affect intergenerational income mobility following the expansion of higher education, how heterogeneous the effects are across regions and between urban and rural areas, and what role different types of education play in reducing intergenerational income correlations and interregional income gaps following the expansion of higher education. This paper aims to explore these issues.

3. Design and methods

Intergenerational mobility refers to whether the income or social status of children has changed relative to that of their fathers' generation. It can be further subdivided into relative intergenerational mobility and absolute intergenerational mobility, which reflect intergenerational mobility from different perspectives. Relative intergenerational mobility reflects changes in the social class status of children relative to the social class status of their fathers. The coefficient of correlation between the income or social status of children and fathers is generally obtained by regressing the income of children and fathers, also known as the intergenerational elasticity coefficient. Greater intergenerational elasticity reflects a strong correlation between the income or class status of children and fathers. The greater the elasticity is, the stronger the correlation between the income or class status of children and fathers and the lower intergenerational mobility is and vice versa. Absolute intergenerational mobility can also be referred to as intergenerational upward mobility, which Chetty et al. (2014) defined as the extent to which the income of a child exceeds the income of his or her father or what the expected level of economic income of children is for a given level of income of their fathers. Children can be considered to have achieved absolute intergenerational mobility if their income or social status exceeds that of their parents in an absolute sense and vice versa. In this work, intergenerational mobility refers mainly to relative intergenerational mobility, which is about judging the change in the class status of children and their fathers in a relative sense rather than whether children surpass their fathers in an absolute sense. It emphasizes how likely it is that children will be able to escape the influence of their fathers' class status and achieve a class leap.

3.1. Measurement of intergenerational mobility

Common measures of intergenerational income mobility include the intergenerational elasticity index and the intergenerational order correlation index. Considering that the real relationship between parental income and children's income may not be linearly related and that the intergenerational elasticity index may be particularly sensitive to samples with low or zero income, obtaining stable results using the log-log intergenerational elasticity index for measurement may prove challenging. Generational order correlation compensates for the limitations of intergenerational elasticity. As proposed by Dahl and DeLeire (2008), generational order correlation refers to the correlation between a parent's position in the distribution of permanent income among all fathers and a child's position in the distribution of permanent income among all children. This correlation reflects the likelihood of a child inheriting his or her parent's income rank order and offers the advantage of robust statistical strength while avoiding issues related to life-cycle bias and temporary income shocks (Chetty et al., 2014; Emran and Shilpi, 2019). Therefore, this paper opts to use intergenerational income order correlation as the measure of intergenerational income mobility.

$$Rank_{ijt}^c = \beta_0 + \beta_{j,t} Rank_i^p + \varepsilon_{ijt} \quad (1)$$

where i , j , and t denote the individual, home province, and birth cohort, respectively. $Rank^c$ and $Rank^p$ represent the permanent income rankings of children and parents in each cohort, respectively. β_0 is a constant

term. $\beta_{j,t}$ represents the correlation between generations of ranking, that is, the intergenerational income index, and ε_{ijt} is the random error term.

3.2. Model setting

To examine the effects of higher education expansion policies on intergenerational mobility, we set the following model specifications:

$$\text{Rank}_{ijt}^c = \beta_0 + \beta_1 \text{Rank}_i^p + \beta_2 \text{intensity}_j \times \text{cohort}_i \times \text{Rank}_i^p + \beta_3 \text{intensity}_j \times \text{cohort}_i + \beta_4 X_{ijt} + \mu_j + \eta_t + \varepsilon_{ijt} \quad (2)$$

In the equation above, i, j, t , Rank^c and Rank^p are the same as in Eq. (1). cohort_i is a dummy variable based on the individual's year of birth and indicates whether the individual was affected by higher education expansion policies. intensity_j indicates the intensity of the higher education expansion of province j . β_0 is the constant term, and X_{ijt} is the control variable, including the type of household of the parents and children, the number of siblings of the children, the compulsory education policy and other control variables. ε_{ijt} is the random error term. μ_j and η_t are provincial fixed effects and birth year fixed effects, respectively; the former controls for characteristics associated with the expansion policy and household size that do not vary with birth cohort, and the latter controls for individual differences in cohorts common to all provinces.

3.3. Data and variables

This study uses data from the Chinese Family Panel Studies (CFPS) from 2010 to 2018. The CFPS is a nationwide, large-scale, multidisciplinary social tracking survey project implemented by the China Social Science Survey Centre of Peking University. The target population of the CFPS is all household members. The data have been officially accessible since 2010. All baseline household members who participated in the 2010 baseline survey and their future blood/adopted children are considered genetic members of the CFPS and are permanently tracked. The database not only contains variables such as persistent income data for children and fathers, education, household registration, and the number of siblings needed for the empirical analysis of this paper but also covers outworkers and information on both cohabiting and non-cohabiting member data, which is convenient for addressing sample selection bias.⁴

3.3.1. Intensity of higher education expansion (intensity)

On 13 January 1999, China's State Council approved and forwarded the Action Plan for the Revitalization of Education for the Twenty-first Century proposed by the Ministry of Education to formally launch a nationwide reform of higher education expansion. The higher education expansion policy was a nationwide education reform, but the degree of expansion varied from province to province, with provinces with richer higher education resources benefiting more from the expansion policy in terms of the number of regular high school graduates. Therefore, the number of college students enrolled before the expansion of higher education is often used to measure the degree of provincial expansion (Lu and Zhang, 2019). In this paper, the ratio of the number of students enrolled in local universities to the number of students enrolled in senior high schools in each province in 1998 was chosen to measure the intensity of the higher education expansion policy. This is mainly based on

the following considerations. First, the main beneficiaries of higher education expansion policies are the students enrolled in senior high schools. The number of students enrolled in senior high schools can predict the number of students enrolled in the entrance examination and affects the number of students eventually recruited into higher education. Therefore, it is necessary to use the number of students enrolled in senior high schools as the basis. Second, after the implementation of China's higher education expansion policy, local colleges and universities were the main bearers of the expansion policy, and new opportunities after the expansion of higher education were attributed to local ordinary colleges and universities. The greater the number of colleges and universities, the greater the number of opportunities for higher education enrolment.⁵ Data show that after the expansion of higher education in 1999, the enrolment of higher education undergraduate students increased rapidly from 1596,800 in 1999 to 10,013,200 in 2021, and the admission rate of higher education increased from 33.86 % in 1998 to 92.89 % in 2021 (see Fig. 1). However, the enrolment of 985 colleges and universities remained stable at approximately 150,000–180,000 students. The admission rates of 985 colleges and universities from 2017 to 2021 were 2.10 %, 2.09 %, 2.01 %, 1.90 %, and 2.30 %, respectively. Therefore, choosing the ratio of the number of students enrolled in local colleges and universities to the number of students enrolled in general high schools in each province is appropriate for measuring the intensity of policies aimed at expanding higher education.

3.3.2. Birth Cohort (cohort)

This paper takes the 1999 implementation of the higher education expansion policy as a pivotal moment in China's education expansion reforms.⁶ The legal age for admission to school is 7 years, and for universities, it is 19 years. Therefore, the individuals in the post-1980 birth cohort were significantly affected by and were the real beneficiaries of higher education expansion policies; they are defined as $\text{cohort}=1$. For individuals born in 1980 or earlier, it is assumed that they were not substantially affected by the higher education expansion policy; therefore, they are defined as $\text{cohort}=0$.

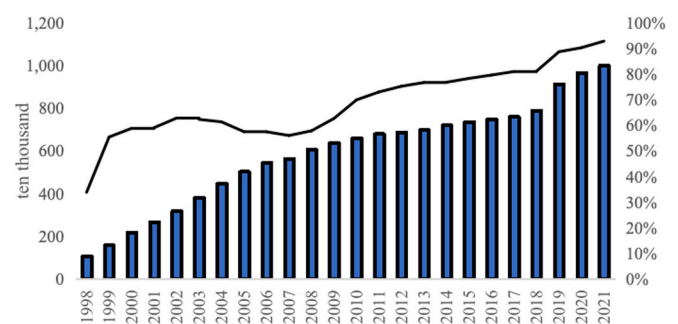


Fig. 1. Enrollment for undergraduate students and admission rate in higher education from 1998 to 2021. Notes: Data from the National Statistical Bulletin on the Development of Education in China.

⁴ Among existing databases, both the CHNS database and the CHIP database contain information on the income of children and parents, but the CHIP database does not use tracking data, and persistent income data for children or parents are not available. Although the CHNS database uses tracking data, the matching data for children and parents are very limited, and the CHNS database does not have relevant data on the types of university institutions of children. It therefore cannot be the primary database for this paper.

⁵ According to affiliation, China's institutions of higher education can be divided into central ministry-affiliated universities, local ordinary colleges, and central-local joint colleges. According to quality level, Chinese higher education institutions can be divided into key and ordinary colleges and universities colleges and universities.

⁶ Cohort is a dichotomous variable including pre-1980 birth cohort and post-1980 birth cohort, where pre-1980 birth cohort includes 1940, 1950, 1960, and 1970 birth cohorts and post-1980 birth cohort includes 1980 and 1990 birth cohorts. The parents' birth cohort is the 1940–1970 birth cohort and the children's birth cohort is the 1970–1990 birth cohort.

3.3.3. Children's Income (Y^C)/Parent's Income (Y^P)

We use the question on income from the CFPS survey data: "How much did you earn in the last 12 months?" Summing the individual incomes yields data on the annual incomes of the child and parent generations.⁷ Incomes for 2012, 2014, 2016 and 2018 are based on the consumer price index adjusted to 2010 price levels. During the matching process, we measured parental income by the father's income level and used the sum of the father's and mother's household income as a proxy variable for parental income.

3.3.4. Child generation income sorting ($Rank^C$)/father generation income sorting ($Rank^P$)

The child's and parent's income rankings are obtained by placing the child's and parent's incomes in their respective birth cohorts after error processing to obtain their income rankings. This study selected as control variables the type of household registration of the children and parents,⁸ the age polynomial of the children and parents, the compulsory education policy,⁹ and the number of siblings of the children. The descriptive statistics of the variables are shown in Table 2.

3.4. Error treatment

When parents and children live together in the same household, both parents' and children's incomes are recorded in the CFPS survey, but there is a problem of missing individual income data in the CFPS database. This includes the following two cases. First, parents and children live together, but at least one of them works temporarily in a place other than the place where their household is located. In this case, the income of the member who has temporarily moved is not recorded in the CFPS survey. Second, when parents and children live in different households, the CFPS survey collects income information for the households in

Table 2
Variables descriptive statistics.

Variable	Obs	Mean	Std. Dev.	Min	Max
$\ln Y^C$	28,202	10.03	0.30	9.18	11.19
$\ln Y^P$	28,202	9.75	0.32	8.68	13.51
$Rank^C$	28,202	0.50	0.29	0.00	1
$Rank^P$	28,202	0.50	0.29	0.00	1
<i>intensity</i>	28,202	0.37	0.15	0.22	1.46
<i>cohort</i>	28,202	0.62	0.49	0.00	1.00
<i>compulsory</i>	28,202	0.85	0.34	0	1
<i>agr_c_rural</i>	28,202	0.68	0.47	0.00	1.00
<i>agr_f_rural</i>	28,202	0.77	0.42	0.00	1.00
<i>age_c</i>	28,202	27.33	6.41	16.00	40.00
<i>age_f</i>	28,202	54.18	6.77	31.00	64.00
<i>sib</i>	28,202	1.45	1.27	0.00	8.00
<i>gender</i>	28,202	0.48	0.50	0.00	1.00
<i>ordinary</i>	8452	0.02	0.14	0.00	1.00
<i>elite</i>	3627	0.05	0.22	0.00	1.00
<i>master</i>	3624	0.03	0.17	0.00	1.00

⁷ Total income refers to the sum of all income received by the household, including net agricultural income, profits from self-employment or private business, government subsidies and benefits, social benefits, pension income, wage income of all household members, income from rental and sales of real estate, interest on savings and income from financial investments, etc.

⁸ 1 indicates rural household, 0 indicates urban household.

⁹ China enacted the Compulsory Education Law in 1986, which implements nine years of compulsory education, although the timing of implementation varies slightly from province to province. The legal age for enrolment in primary school is 7, and compulsory education ends at 15. Subtracting 15 years from the number of years in which the Compulsory Education Law is implemented in each province is the borderline number of years affected by the Compulsory Education Law.

which the parents or children live rather than for the respective households at the same time, and the incomes of members of the non-surveyed households are not collected. If the first scenario is used for empirical estimation, the missing income data are likely to introduce a sample truncation problem, which may have different effects on inter-generational mobility across birth cohorts before and after the expansion of higher education. For example, birth cohorts of different ages have different probabilities of children living with their parents or experiencing temporary migration, which directly affects whether individual income information is captured by the CFPS survey. Therefore, we believe that the sample truncation problem is different for each birth cohort before and after the expansion of higher education, but the unique characteristics of the CFPS survey data allow us to address this sample selection bias. The CFPS survey data collect information on education, employment status, and occupation of noncohabiting and temporarily relocated household members and relatives of each household member.¹⁰ This provides support for this paper to use CFPS survey data to correct sample selection bias and predict income.

3.4.1. Life-cycle bias and temporary income shocks

Life-cycle bias refers to the difference in the impact of income at different ages of the father's generation on the income of the children at different ages. For example, the impact of the father's income on the income of the children at age 50 or 40 is different, and using the income of the father's generation or the children's generation when they are fixed or too young in age may imply attenuation bias. In this study, following the treatment of Zou and Ma (2019), the age limits of fathers and children are strictly restricted by excluding samples with children under 16 years of age and fathers over 64 years of age and removing samples with an age difference between fathers and children under 14 years of age and school students.

Temporary income shocks refer to the systematic bias between an individual's one-year income and his or her permanent income. If intergenerational income mobility is measured in terms of a given year's income, it may include temporary income shocks with measurement error, which can lead to an upwards or downwards bias in the estimation of intergenerational income elasticities. For this partial bias, this paper uses the 2010–2018 CFPS data to select the average of at least two periods of individual respondent income in the five survey datasets as a proxy for permanent income. Additionally, polynomials in the age of fathers and children are added, and the average of the father's and mother's educational attainment is used as an instrumental variable to estimate the lifetime income of the father's generation (Fan et al., 2021).

3.4.2. Selection bias

Most surveys routinely interview family members living together, such as parents and children, who share a common household identification number. However, children may leave their original family through marriage or other reasons and form new families, which generates new household identification numbers. This can lead to a large amount of missing data, as parents and children cannot be fully matched. Furthermore, it can increase measurement error in intergenerational income mobility. In the intergenerational economic analysis framework developed by Becker and Tomes (1986), which includes parental human capital investment, family cultural capital is an important factor that influences intergenerational mobility. The impact of family cultural capital is proportional to the amount of time parents and children live together. Ignoring the co-residence bias may lead to overestimation of the estimation results. Moreover, the 2000 census data show that the number of Chinese nuclear families is gradually increasing and that the proportion of parents and children who live together is decreasing. Therefore, it is important to investigate the effect of co-residence bias. In the actual data selection and matching process,

¹⁰ Including spouses, immediate family members, and siblings.

data must be processed to obtain samples that include co-residence and non-co-residence data.

Migration bias has a similar effect as co-residence bias, but it mainly refers to family members who work outside their home for a temporary period instead of living there all year. Their income may still be a significant part of their household income, but some databases may not account for this group. For example, the CFPS database does not include income information for temporary migrants. Nevertheless, in recent years, China has undergone significant socialist market economy reforms, resulting in a significant increase in the number and speed of temporary labour mobility. The income of temporary migrants has also increased significantly compared to the past. Ignoring the income information of temporary migrants may lead to serious sample selection bias. Therefore, in addition to addressing co-residence bias, this study addresses the potential estimation bias caused by temporary migration bias. To reduce this estimation bias, this paper adopts the data processing method of Fan et al. (2021) and employs the Heckman sample selection model to correct and predict missing data on parent or child income. Specifically, this study first estimates the sample of adult children, which includes samples with recorded income as well as missing income data, and sets the probit model as follows:

$$W_{ipt} = \beta_0 + \beta_1 m_{ipt} + \beta_2 X_{ipt} + \varepsilon_{ipt} \quad (3)$$

where W_{ipt} is a dummy variable for whether the child's income information is recorded in the questionnaire. It is defined as 1 if the child's income is recorded in the questionnaire and 0 if not. m_{ipt} is an exclusion constraint variable for the number of children's siblings. X_{ipt} is a dummy variable that includes individual years of education, parental years of education, a polynomial of age, gender, whether the individual lives in a coastal area, and household registration. ε_{ipt} is the term of random error. When the Heckman sample selection model is used for analysis, the probit model in the first stage is used to test whether income information has been recorded. The second-stage model is the income correction model, which calculates the inverse Mills ratio and uses it as a control variable in the second-stage equation to correct and predict individual income:

$$\ln Y_{ipt} = \beta_3 + \beta_4 X_{ipt} + \beta_5 \lambda_i + \varepsilon_{ipt} \quad (4)$$

Y_{ipt} is the individual income recorded in the database, X_{ipt} is the relevant control variable (same as Eq. (3)), λ_i is the inverse Mills ratio, and ε_{ipt} is the random error term.

3.5. Data sources

The China Family Panel Studies (CFPS) 2010–2018 are the primary source for this paper based on the following considerations. First, the CFPS database covers the distribution of key demographic and socioeconomic variables needed for the empirical analyses in this paper, including data on age, gender, schooling, and urban–rural stratification, which are similar to Chinese census data. Second, the CFPS database covers urban and rural areas in 25 provinces, municipalities, and autonomous regions in China, representing 95 % of China's population. In addition, the CFPS survey is a nationwide, large-scale, multidisciplinary social tracking survey project with data that have been formally accessible since 2010. All baseline household members and their blood/adopted children who participated in the 2010 baseline survey are considered genetic members of the CFPS and are tracked in perpetuity, similar to the Panel Study of Income Dynamics in the United States.

4. Empirical analysis

4.1. Effects of intergenerational income mobility of higher education expansion

4.1.1. Estimation of intergenerational mobility effects of higher education expansion

Table 3 presents the results of the empirical estimation of the effect of higher education expansion on intergenerational income mobility and educational mobility. The results in Columns (1)–(3) show that the coefficients of $Intensity \times cohort \times Rank^p$ are significantly positive in all three models, suggesting that higher education expansion may have increased the income correlation between children and parents and reduced intergenerational mobility. The likelihood that children are in the same income bracket as their parents increased following the expansion of higher education.

Given the nature of the college expansion policy, its impact on educational mobility could be more direct and pronounced than on the impact on income mobility. We further examined the intergenerational educational mobility effects of higher education expansion policies. The results are presented in Columns (4) to (7) and show that the estimated coefficients of $Intensity \times Cohort \times Edu^p$ are significantly positive in Columns (4) to (5), whereas the estimated coefficients of the interaction term do not pass the significance test in Columns (6)–(7). This suggests that the implementation of a higher education expansion policy significantly reduces intergenerational mobility in education and fails to significantly improve intergenerational mobility in higher education. This may be related to the key role of family background in intergenerational educational mobility. At the elite higher education stage, China's higher education resources are scarce, and higher education opportunities are prioritized for the offspring of families with higher academic qualifications or higher income levels. After entering the stage of the popularization of higher education, capital advantaged families continue to lead in the possession of higher education resources and are more likely to be beneficiaries of higher education expansion policies, while the offspring of capital disadvantaged families are excluded from the threshold of quality education resources (Guo et al., 2019). These results further support the robustness of the conclusion that the university expansion policy does not significantly increase intergenerational income mobility.

4.1.2. Instrumental variable test

We use the instrumental variable method as the primary solution to address potential endogeneity problems to ensure the reliability and robustness of the research findings. We choose the number of local colleges and universities in 1998 as an instrumental variable. The variable choice is based on the following considerations:

First, the number of local colleges and universities in 1998 significantly affected the intensity of higher education expansion in 1999. The *Action Plan for Revitalizing Education for the 21st Century* states that "in order to give more high school graduates the opportunity to pursue higher education, the total number of students enrolled in local higher education will reach 6.6 million in the year 2000, based on local demand and the possibilities of financial investment and teaching staff, and on the premise of adopting a new mechanism and model". This document reveals that the expansion of colleges and universities is aimed mainly at increasing the enrolment rate of local general colleges and universities, although a higher education expansion policy has been initiated for the whole country. However, due to the differences in higher education resources across provinces in China, the intensity of higher education expansion differs significantly across provinces. In provinces where higher education resources are more abundant, it is easier for ordinary high school graduates to gain access to higher education.

Second, using the number of local colleges and universities in 1998 as an instrumental variable does not directly affect the income rank of children. Furthermore, it is not directly related to the random error

Table 3
Estimation results of intergenerational mobility effects of higher education expansion.

Variable	(1) <i>Rank^c</i>	(2) <i>Rank^c</i>	(3) <i>Rank^c</i>	(4) <i>Edu^c</i>	(5) <i>Edu^c</i>	(6) <i>Cedu^c</i>	(7) <i>Cedu^c</i>
<i>Edu^p</i>				0.572*** (44.13)	0.491*** (26.51)		
<i>Cedu^p</i>						0.0749 (1.366)	−0.0829 (−0.719)
<i>Rank^p</i>	0.379*** (33.49)	0.308*** (23.78)	0.164*** (17.87)				
<i>intensity × cohort × Edu^p</i>				0.245*** (6.973)	0.126*** (2.912)		
<i>intensity × cohort × Cedu^p</i>						0.0536 (0.449)	0.0241 (0.192)
<i>intensity × cohort</i>	−0.0749*** (−3.227)	−0.148*** (−3.653)	−0.106*** (−3.929)	−0.0416 (−0.371)	−0.715*** (−3.815)	−0.403 (−0.604)	−0.257 (−0.368)
<i>intensity × cohort × Rank^p</i>	0.277*** (8.373)	0.122*** (3.196)	0.0539** (2.021)				
<i>control variables</i>	No	No	Yes	No	Yes	No	Yes
<i>birth year fixed effects</i>	No	Yes	Yes	No	Yes	No	Yes
<i>province fixed effects</i>	No	Yes	Yes	No	Yes	No	Yes
<i>Constants</i>	0.284*** (41.99)	0.500*** (18.34)	−0.863 (−0.728)	1.641*** (49.51)	2.831*** (13.73)	5.538*** (13.11)	13.39 (1.554)
<i>Observations</i>	28,202	28,202	28,202	28,202	28,202	409	409
<i>R²</i>	0.183	0.257	0.712	0.194	0.258	0.194	0.217

Note: 1. *Edu^c* and *Edu^p* indicate the final educational level of children and parent, respectively, with 1, 2, 3, 4 and 5 being defined for illiteracy, primary, junior, junior high and higher education respectively.

2. *Cedu^c* and *Cedu^p* indicate whether children and parents have completed tertiary education, and are defined as 1 if they have and 0 otherwise.

3. ***, **, * indicate significant at the 1 %, 5 % and 10 % levels respectively; data in brackets are t-statistics. 4. Given that the literature has examined the intergenerational educational mobility effects of higher education expansion policies, this paper focuses on the intergenerational income mobility effects of higher education expansion policies, so there is no further discussion of the intergenerational educational mobility effects later.

term, which is strictly exogenous, so the number of local colleges and universities in 1998 is a better instrumental variable. Following the treatment of Nunn and Qian (2014), this study constructs the interaction term of this instrumental variable with the birth cohort and parents' income percentile rank as an instrumental variable to examine the impact of higher education expansion policies on intergenerational income (see Table 4).

The Cragg–Donald Wald F and Kleibergen–Paap Wald F statistic values are significantly greater than 16.39, which are Stock–Yogo weak

Table 4
Estimation results of Instrumental variable.

Variable	IV I		IV II <i>Rank^c</i>
	<i>intensity * cohort</i>	<i>intensity * cohort * Rank^p</i>	
<i>iv × cohort</i>	−0.00254*** (−13.46)	−0.00372*** (−22.95)	
<i>iv × cohort × Rank^p</i>	0.00315*** (28.74)	0.00875*** (93.18)	
			0.126*** (11.05)
<i>intensity × cohort</i>			−1.322*** (−8.321)
<i>intensity × cohort × Rank^p</i>			0.475*** (6.094)
<i>control variables</i>	Yes	Yes	Yes
<i>birth year fixed effects</i>	Yes	Yes	Yes
<i>province fixed effects</i>	Yes	Yes	Yes
<i>Constants</i>	0.388 (0.820)	0.0935 (0.231)	−0.564 (−0.455)
<i>Observations</i>	28,202	28,202	28,202
<i>R²</i>	0.894	0.868	0.660
<i>CD Wald F</i>			353.073
<i>KP-LM</i>			181.010*** (0.0000)
<i>KP-Wald F</i>			106.374

Note: ***, **, * indicate significant at the 1 %, 5 % and 10 % levels respectively; data in brackets are t-statistics.

ID test critical values at the 10 % level (Stock and Yogo, 2002), indicating that there is no problem of weak instrumental variables and that the instrumental variables are valid. In addition, the estimated coefficient of the instrumental variable interaction term is significantly positive, which is consistent with the estimation of the baseline model.

4.1.3. Robustness tests

In this work, robustness tests are performed by replacing variables and randomly intercepting samples. The results are shown in Table 5.

In Column (1), the intensity of the higher education expansion policy is replaced by the ratio of the number of university enrolments in 1998 to the household population in 1998. In Column (2), the parental income percentile rank is replaced by the household income percentile rank. In Column (3), the parental income percentile rank is replaced by the parental sorting position obtained by sorting fathers in the children's birth cohort. In Model 4, the age of the sample is strictly restricted to the age range from 20 to 60. The results show that the estimated coefficients of *intensity × cohort × Rank^p* in Table 5 are all significantly positive, suggesting that the income level of children after the expansion of higher education has been strengthened by the influence of parents' income class and that intergenerational persistence has increased, which is consistent with the previous estimation results.

Given the possibility that there may have been other policies implemented during the same period that affected intergenerational income mobility, ignoring these potential effects would also lead to biased estimates. There are two main treatments used in this paper: the first is the randomization of the treatment group, and the second is the randomization of the implementation time of the policy.

For the first treatment, we refer mainly to the treatment of Chen et al. (2020), which sets the 1970–1975 birth cohort as the control group and the 1976–1980 birth cohort as the treatment group before the implementation of the higher education expansion policy (Model 5). After the implementation of the higher education expansion policy, we set the 1980–1986 birth cohort as the control group and the 1987–1994 birth cohort as the treatment group (Model 6). The results of Columns (5) to (6) in Table 5 show that the estimated coefficients of the interaction

Table 6

Heterogeneity estimation results of Urban-rural and regional.

variable	Model 1 Rural area <i>Rank</i> ^c	Model 2 Urban area <i>Rank</i> ^c	Model 3 Eastern region <i>Rank</i> ^c	Model 4 Middle Region <i>Rank</i> ^c	Model 5 Western area <i>Rank</i> ^c
<i>intensity</i> × <i>cohort</i> × <i>Rank</i> ^p	−0.0655 (−1.419)	0.162*** (4.521)	0.108*** (2.818)	−0.142*** (−2.808)	−0.186*** (−2.802)
<i>intensity</i> × <i>cohort</i>	−0.0545 (−1.411)	−0.139*** (−3.776)	−0.0899** (−2.464)	−0.0689 (−1.203)	0.0195 (0.270)
<i>Rank</i> ^p	0.118*** (11.05)	0.198*** (10.63)	0.149*** (6.841)	0.146*** (10.21)	0.155*** (10.76)
control variables	Yes	Yes	Yes	Yes	Yes
birth year fixed effects	Yes	Yes	Yes	Yes	Yes
province fixed effects	Yes	Yes	Yes	Yes	Yes
Constants	0.732 (0.600)	−3.791*** (−2.946)	0.763 (0.414)	−5.240*** (−3.169)	−0.383 (−0.194)
Observations	17,287	10,915	11,378	8935	7889
<i>R</i> ²	0.694	0.663	0.669	0.712	0.667

Note: ***, **, * indicate significant at the 1 %, 5 % and 10 % levels respectively; data in brackets are t-statistics.

Table 5

Estimation results of robustness tests.

Variable	(1) <i>Rank</i> ^c	(2) <i>Rank</i> ^c	(3) <i>Rank</i> ^c	(4) <i>Rank</i> ^c	(5) <i>Rank</i> ^c	(6) <i>Rank</i> ^c
<i>intensity</i> × <i>cohort</i> × <i>Rank</i> ^p	0.287*** (4.055)	0.0741*** (2.914)	0.0766*** (2.961)	0.0539** (2.021)		
<i>intensity</i> × <i>cohort</i>	−0.362*** (−5.606)	−0.123*** (−4.645)	−0.128*** (−4.849)	−0.106*** (−3.929)		
<i>Rank</i> ^p	0.163*** (17.82)	0.154*** (17.48)	0.154*** (16.70)	0.164*** (17.87)	0.228*** (11.44)	0.220*** (14.33)
<i>intensity</i> × <i>cohort</i> (1970–1980) × <i>Rank</i> ^p					−0.0286 (−0.971)	
<i>intensity</i> × <i>cohort</i> (1981–1994) × <i>Rank</i> ^p						0.00704 (0.153)
<i>intensity</i> × <i>cohort</i> (1970–1980)					−0.00125 (−0.0409)	
<i>intensity</i> × <i>cohort</i> (1981–1994)						−0.130*** (−2.984)
control variables	Yes	Yes	Yes	Yes	Yes	Yes
birth year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
province fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Constants	−0.896 (−0.759)	−0.607 (−0.482)	−1.058 (−0.854)	−0.863 (−0.728)	3.171* (1.900)	0.476*** (12.42)
Observations	28,202	28,202	28,202	28,202	17,314	10,888
<i>R</i> ²	0.712	0.717	0.712	0.712	0.692	0.646

Note: ***, **, * indicate significant at the 1 %, 5 % and 10 % levels respectively; data in brackets are t-statistics.

terms do not pass the significance test for either the 1970–1980 birth cohort or the 1981–1994 birth cohort, suggesting that the results are robust.

For the second treatment, we randomly switch the values of the higher education expansion policy intensity in 25 provinces to randomize the timing of the policy and re-estimates the interaction terms of the policy intensity with the birth cohort and the percentile ranking of the parents' income. We find that the coefficients of the interaction terms obtained after the switch approximately follow the normal distribution centred on 0 after the 500 repeated replacement tests, again indicating that the estimation results are robust (see Fig. 2).

4.2. Heterogeneity analysis of the impact of higher education expansion on intergenerational mobility

While higher education expansion policies have significantly increased overall higher education enrolment opportunities, the level of higher education resources may vary across regions in China either as a result of historical constraints or the level of economic development, which means that access to higher education varies across regions. For economically developed regions such as urban areas or regions with a relative advantage in higher education resources, the number of places

allocated to higher institutions may be relatively higher and the number of senior secondary school graduates enrolled may be correspondingly higher. Thus, it is necessary to further investigate whether there is urban–rural and regional heterogeneity in the impact of higher education expansion on intergenerational income mobility.

First, the results show that the estimated coefficient of *intensity* * *cohort* * *Rank*_{pi} for rural areas does not pass the significance test. The estimated coefficient of *intensity* * *cohort* * *Rank*_{pi} for urban areas is significantly positive, suggesting that higher education expansion policies significantly reduced intergenerational mobility in urban areas and that intergenerational inheritance may have increased in urban areas after the expansion, whereas the effect on intergenerational mobility in rural areas was more limited.

Second, the results show that the estimated coefficient of *intensity* * *cohort* * *Rank*_{pi} is significantly positive for the eastern region and negative for the central and western regions, suggesting that the expansion of higher education increased intergenerational mobility more in the less developed regions, such as the central and western regions, and the pull effect may have been more limited for the developed eastern regions. That is, while the expansion of higher education reinforces intergenerational income transmission in relatively developed regions, it also increases intergenerational mobility in relatively less

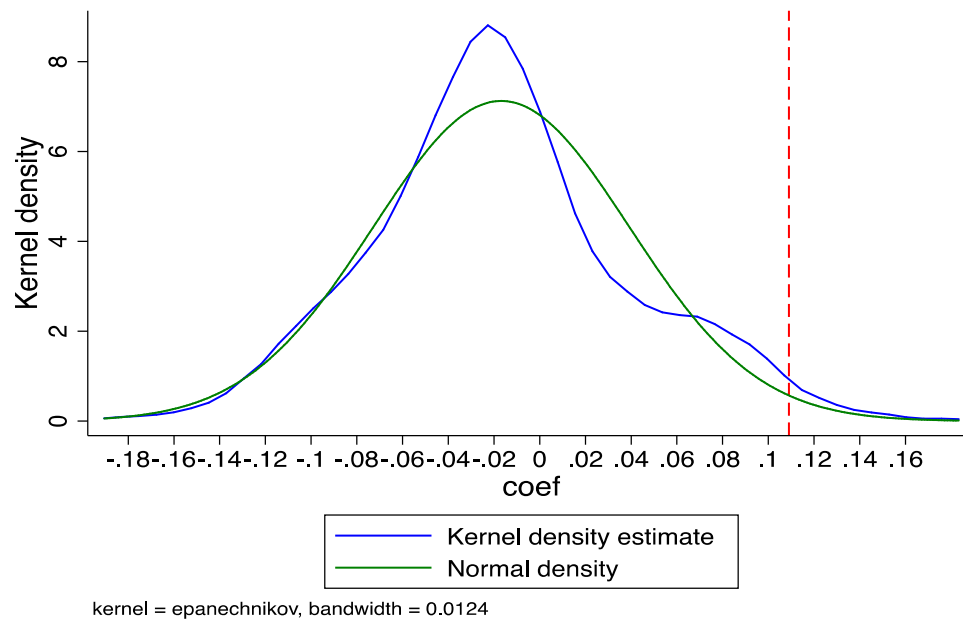


Fig. 2. Robustness test.

developed regions. It therefore becomes possible for children in less developed regions to move out of their parents' income brackets and achieve intergenerational mobility.

In addition, this study draws on the classification method of Zhang and Li (2019) and uses the ratio of the number of regional higher education admissions to the number of regional six-year primary school graduates to calculate overall regional access. The Thief index method is used to estimate the degree of regional access differences by subdividing administrative regions into regions with absolute advantages, relative advantages, and relative disadvantages in higher education opportunities.¹¹ Furthermore, the regional higher education resource abundance level is classified according to the university competitiveness scores of each province and its rankings. This study considers provinces whose rankings are in the top 10 to be higher education resource-rich provinces; otherwise, they are considered higher education resource-inexpensive provinces. The detailed estimation results of the higher education resource abundance level are shown in Table 7.

The results show that the estimated coefficients of $intensity * cohort * Rank_{pi}$ of Models 1 and 2 in Table 7 are 0.175 and -0.072 , respectively, and pass the significance test, whereas the estimated coefficient of Model 3 is 0.001 but does not pass the significance test. This suggests that the increase in access to higher education after the expansion of higher education reinforces the phenomenon of intergenerational income inheritance in areas with absolute superiority in higher education resources but significantly improves intergenerational income mobility in areas with relative superiority in higher education resources. However, the improvement effect of intergenerational income mobility is not good for relatively disadvantaged districts in terms of higher education resources. Models 4 and 5 in Table 7 show the estimation results for educational resource-rich and educational resource-poor provinces, respectively. The estimated coefficients of the core interaction terms are significantly positive for the former and the core interaction terms are significantly negative for the latter, suggesting that the higher education

resource-rich provinces exhibit significant intergenerational income inheritance following the expansion of higher education, with children's dependence on their father's income bracket significantly strengthened. Unfortunately, the higher education resource-poor provinces do not significantly improve intergenerational income mobility.

The results of the overall estimation of heterogeneity basically verify the preliminary conjecture of the paper that the expansion of higher education has increased intergenerational mobility in less economically developed regions, such as rural or central and western regions. This is associated with the State's vigorous support for the development of western regions, the implementation of the counterpart support plan and the vigorous promotion of the implementation of the strategy of the war against poverty in recent years⁴. Furthermore, with the expansion of the scale of higher education, Chinese higher education has also developed a relatively comprehensive system of educational assistance, including student loans, scholarships, and state educational subsidies for military service. All of these have, to some extent, alleviated the financial burden imposed on poor families through the marketisation of higher education, prevented the children of poor families from losing the opportunity for higher education due to family financial constraints, and theoretically given them the possibility of changing their fortunes and achieving class mobility. However, in areas where higher education resources are relatively disadvantaged or where quality education resources are not abundant, the pulling effect of higher education expansion is relatively limited. The strong policy push does not significantly improve educational inequality in underresourced areas or significantly improve intergenerational mobility and may even reinforce intergenerational income inheritance in resource-rich or economically developed areas. This may be one of the reasons why social phenomena such as the "uselessness of studying" and the "devaluation of diplomas" are widely discussed in society.

4.3. Further discussion

Considering that higher education expansion significantly affects access and income levels at different levels of quality and qualification in higher education institutions, with reference to Jia et al. (2022), this paper further explores how elite higher education as well as higher levels of postgraduate education affect intergenerational mobility after higher education expansion. The results in Table 8, Panels C to E, show that higher education, elite higher education and postgraduate

¹¹ Areas of absolute advantage include Beijing, Tianjin, and Guangdong. Areas of relative advantage include Inner Mongolia, Liaoning, Jilin, Heilongjiang, Zhejiang, Fujian, Shandong, Hubei, Hunan, Chongqing, Shaanxi, Gansu, Qinghai, and Ningxia. Areas of relative disadvantage include Hebei, Shanxi, Jiangsu, Jiangxi, Henan, Guangdong, Guangxi, Hainan, Sichuan, Guizhou, Yunnan, Tibet, and Xinjiang.

Table 7

Heterogeneity estimation results of higher education resources.

Variable	Model 1 Areas of absolute advantage <i>Rank</i> ^c	Model 2 Areas of relative advantage <i>Rank</i> ^c	Model 3 Areas of relative disadvantage <i>Rank</i> ^c	Model 4 Resource-rich provinces <i>Rank</i> ^c	Model 5 Resource-poor provinces <i>Rank</i> ^c
<i>intensity</i> × <i>cohort</i> × <i>Rank</i> ^p	0.175*** (2.891)	−0.0772* (−1.656)	0.001 (0.00814)	0.0748** (2.033)	−0.0462 (−1.115)
<i>intensity</i> × <i>cohort</i>	−0.146* (−1.918)	−0.107** (−2.353)	0.271** (2.427)	−0.0597* (−1.653)	−0.0127 (−0.251)
<i>Rank</i> ^p	0.181*** (3.547)	0.171*** (12.41)	0.142*** (10.46)	0.215*** (11.17)	0.159*** (15.34)
<i>control variables</i>	Yes	Yes	Yes	Yes	Yes
<i>birth year fixed effects</i>	Yes	Yes	Yes	Yes	Yes
<i>province fixed effects</i>	Yes	Yes	Yes	Yes	Yes
<i>Constants</i>	−7.036*** (−5.926)	−0.379 (−0.184)	−0.794 (−0.567)	−2.819 (−1.479)	−0.497 (−0.391)
<i>Observations</i>	1917	11,131	15,154	11,076	17,126
<i>R</i> ²	0.626	0.714	0.693	0.695	0.702

Note: ***, **, * indicate significant at the 1 %, 5 % and 10 % levels respectively; data in brackets are t-statistics.

Table 8

Estimation results of further analysis.

VARIABLES	Model 1 Full sample	Model 2 Mid-aged parents	Model 3 1st-married families
Panel A : 10-year grouping	Dependent variable: <i>Rank</i>^c		
<i>Rank</i> ^p	0.438*** (64.66)	0.449*** (48.16)	0.438*** (61.82)
Constant	0.288*** (75.53)	0.280*** (51.13)	0.291*** (72.81)
Observations	17,541	9462	16,177
<i>R</i> ²	0.192	0.197	0.191
Panel B : 5-year grouping	Dependent variable: <i>Rank</i>^c		
<i>Rank</i> ^p	0.451*** (67.48)	0.465*** (51.01)	0.451*** (64.52)
Constant	0.283*** (75.04)	0.278*** (52.49)	0.286*** (72.49)
Observations	17,541	9462	16,177
<i>R</i> ²	0.206	0.216	0.205
Panel C: role of ordinary university	Dependent variable: <i>Rank</i>^c		
<i>Rank</i> ^p	0.213*** (22.14)	0.234*** (18.46)	0.214*** (21.47)
ordinary	0.143*** (7.056)	0.135*** (5.010)	0.147*** (6.896)
Constant	0.599*** (91.70)	0.580*** (65.56)	0.598*** (88.61)
Observations	5746	3407	5420
<i>R</i> ²	0.091	0.101	0.090
Panel D: role of elite university	Dependent variable: <i>Rank</i>^c		
<i>Rank</i> ^p	0.126*** (10.74)	0.144*** (9.005)	0.126*** (10.44)
elite	0.0839*** (5.569)	0.0740*** (3.644)	0.0866*** (5.482)
Constant	0.732*** (88.06)	0.715*** (61.75)	0.732*** (85.38)
Observations	2364	1405	2245
<i>R</i> ²	0.066	0.069	0.065
Panel E: Graduate education	Dependent variable: <i>Rank</i>^c		
<i>Rank</i> ^p	0.131*** (11.08)	0.146*** (9.023)	0.131*** (10.70)
master	0.0498*** (2.590)	0.0271 (1.043)	0.0468** (2.372)
Constant	0.730*** (86.43)	0.715*** (60.61)	0.730*** (83.76)
Observations	2371	1409	2252
<i>R</i> ²	0.054	0.056	0.053

Note: ***, **, * indicate significant at the 1 %, 5 % and 10 % levels respectively; data in brackets are t-statistics.

Note: All results are based on estimates from the post-1980 birth cohort.

education significantly improve child income classification.

Specifically, the baseline intergenerational mobility model yields an estimated coefficient of 0.438 for intergenerational elasticity (Panel A). After considering elite higher education and postgraduate education (Panel D and Panel E), the coefficient of intergenerational income elasticity decreases to 0.126 and 0.131, respectively, indicating that elite higher education and postgraduate education significantly reduce the income correlation between fathers and children and increase intergenerational mobility. In contrast, the effect of general higher education on intergenerational income correlation is significantly weaker (Panel C), and the decrease in the elasticity coefficient is smaller than that of elite higher education and postgraduate education. According to current estimates, elite higher education and postgraduate education significantly improve the income distribution of children and increase intergenerational income mobility. Although elite higher education and postgraduate education have positive effects in terms of reducing intergenerational income correlation and increasing intergenerational mobility, moving the parent from the lowest to the highest 20th percentile of the ranking improves the income ranking of the child by approximately 10 percentage points¹² compared with approximately 8.4 percentage points for elite higher education and approximately 5 percentage points for postgraduate education (Panel D and Panel E).¹³ In other words, while elite individuals with higher and postgraduate education have more improved intergenerational mobility, the intergenerational transmission effects among households are not negligible. All results are insensitive to the use of alternative subsamples, as shown in the remaining columns of the same table. The estimates in this subsection suggest that family background is likely to remain the most important factor that influences the level of intergenerational mobility.

Since the purpose of higher education expansion is to provide a “ticket” to higher education for those who would not otherwise have access to it, the new increase in higher education opportunities is more likely to persist in general rather than elite institutions. Although most school-age groups now have easier access to higher education because of this development, there is still fierce rivalry for admission to elite key universities, making admission to these institutions difficult. However, the massive expansion of higher education in the short term has led to a mismatch between supply and demand in the labour market, which has greatly reduced the functional value of higher education diplomas, especially general higher education diplomas. In contrast, the greater availability of higher education resources has further reduced the distinction between undergraduate and specialist qualifications. In

¹² Calculated formula: $0.126 \times 80 \times 100 \% = 10.08 \%$; $0.131 \times 80 \times 100 \% = 10.81 \%$.

¹³ Formula: $0.0839 \times 100 \% = 8.39 \%$; $0.0498 \times 100 \% = 4.98 \%$.

contrast to the shrinking of the general bachelor's degree and the decline in signalling capacity, elite colleges and universities are still responsible for selecting and training the elite of society, and postgraduate education plays an important role in training high-level and innovative talent. In terms of human capital theory and signalling mechanisms, a higher qualification is a signal for the monopoly of certain job opportunities and access to a higher class, and an increase in relative educational attainment means greater competitiveness in employment and a higher return on investment. Therefore, both postgraduate and elite higher education are among the main ways to promote individual income growth and achieve class mobility.

In addition, this paper reveals that family background continues to play a strong role in determining children's income ranking. Family background can directly influence the ultimate social status of children by influencing their educational opportunities. The relationships among the three factors change with social development. Before the expansion of higher education, family background may have influenced children's access to higher education, but children from disadvantaged backgrounds still had the opportunity to achieve intergenerational mobility through higher education, and access to higher education was essentially the same as obtaining a "ticket" for class mobility. However, the mismatch between supply and demand caused by the expansion of higher education has reduced the signalling value of higher degrees and made predisposing influences such as family background more important.

On the one hand, advantaged families are more likely to acquire parenting styles and practices from a variety of sources and to invest more time in companionship and communication to develop their children's noncognitive skills, which significantly influences their children's employment and income levels later in life (Guryan et al., 2008; Lindqvist and Vestman, 2011). On the other hand, advantaged families also help their children gain an advantage in job search and employment by investing directly in high levels of educational capital or by providing access to high-quality educational resources or elite education (Müller and Pollak, 2004). As the expansion of higher education has been accompanied by rapid economic and social change in China, the resource disparities resulting from families' resource endowments may be translated into further widening of income disparities between different classes of families, which reinforces the "Matthew effect" of intergenerational class reproduction or the transmission of intergenerational poverty. Whether through direct or indirect intergenerational transmission, the intergenerational transmission effect continues to play a strong role in influencing the income class and intergenerational mobility opportunities of children.

4.4. Higher education expansion, intergenerational income flows, and income inequality—a Gatsby Curve

Corak (2013) proposed the "Gatsby curve" to depict the relationship between income inequality and intergenerational income mobility: the higher the level of social inequality in a country or region is, the greater the impact of parental income level on children's income level. Our previous empirical analyses show that the expansion of higher education increases the intergenerational income correlation and reduces intergenerational income mobility. How does the relationship between intergenerational income mobility and income inequality change after the expansion of higher education? Is the increase in social inequality accompanied by a decrease in intergenerational income mobility? Using the Gatsby curve plotting method, this paper plots the changes in the relationship between the intergenerational income anecdote index (rank-rank) and the Gini coefficient at the provincial level.

Corak (2013) argued that children's formative years can be used to

better measure the Gini coefficient of their father's generation's income. This paper therefore chooses the Gini coefficients of 1989 and 2004 as proxies for social inequality in the period before and after the expansion of higher education.¹⁴ The results show that the intergenerational income anecdote index had a significant correlation with the Gini coefficient for the birth cohort before the expansion of higher education (see Fig. 3). In contrast, the Gini coefficient for the postexpansion birth cohort had a significant positive relationship with the intergenerational income anecdote index (see Fig. 4), and the increase in the degree of income inequality was accompanied by an increase in the intergenerational income correlation. These findings suggest that with an increase in income inequality, the descendants of poor families have significantly increased difficulty with upward mobility, and high income inequality after higher education expansion is accompanied by greater intergenerational inequality. The results reveal that there is a Gatsby curve in China, which is consistent with the findings of Fan et al. (2021).

5. Conclusions

China's higher education expansion policy has broadly increased access to higher education across society, but the positive benefits of this policy in improving intergenerational income mobility across society have been insufficient.¹⁵ In particular, the actual policy effects of higher education expansion may be more limited in less developed areas, such as rural or midwestern regions, or in provinces with less abundant higher education resources and opportunities. Both elite higher education and postgraduate education contribute to the increase in children's income status and the decrease in the correlation between generations of income, but we find no evidence of a weakening in the effect of intergenerational family transmission, and family background remains the most important factor that influences children's income status. Following the expansion of higher education, there is a Gatsby curve in China, and an increase in income inequality is accompanied by greater intergenerational inequality.

Currently, the main contradiction in Chinese society is between people's growing need for a better life and unbalanced and insufficient development, which is reflected mainly in the obvious income gap between urban and rural areas and different social strata. As an important mechanism for social stratification, higher education should be a "knocking brick" for high-prestige and high-income occupations. Nevertheless, this study finds that increased access to higher education has not substantially increased income mobility between generations and that society at large is still faced with some degree of income inheritance and poverty transfer between generations. In the long run, as society becomes more competitive, the efficiency and smoothness of social class mobility will decline significantly, which is likely to accelerate social stratification. Over the past two decades, the Chinese government has been committed to promoting the rapid development of higher education, but this development has been in the form of expanding the scale of higher education rather than improving the quality of education. The high-level talent cultivated in higher education often fails to meet the needs of national strategy and regional economic development, resulting in a structural mismatch between higher

¹⁴ Data from China Statistical Yearbook, China Bureau of Statistics, and Gini coefficients of various regions are calculated.

¹⁵ The results of this paper and those of Mok (2016) are fairly comparable in that they both concluded that increasing access to higher education did not greatly increase intergenerational mobility. However, Mok's (2016) study focused on presenting the differences in salary levels and employment rates of college students after higher education expansion in a descriptive form and did not provide an in-depth discussion of the correlation between higher education expansion and intergenerational mobility, and was unable to accurately identify the specific effects of higher education expansion policies on individual class leaps, which is a shortcoming that this paper addresses.

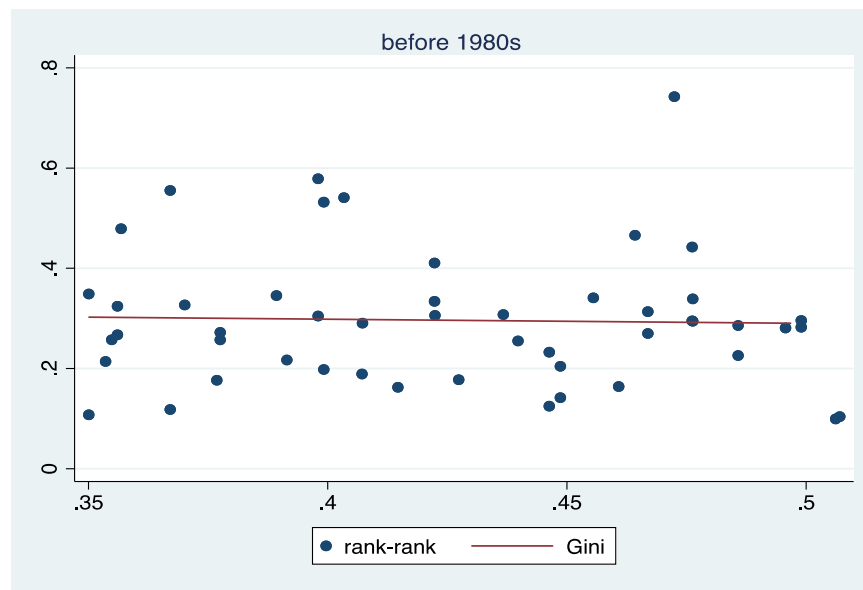


Fig. 3. The Gatsby curve for birth children before 1980s.

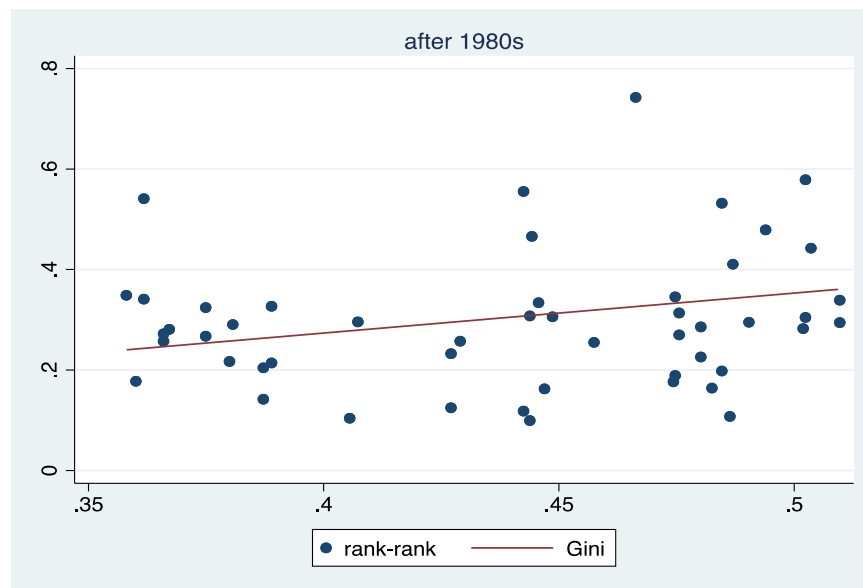


Fig. 4. The Gatsby curve for birth children after 1980s.

education and industrial development. Therefore, promoting the high-quality development of higher education can help to promote intergenerational income mobility and avoid social stratification and solidification. This study reveals that postgraduate education and elite higher education are important mechanisms for reducing intergenerational persistence and improving intergenerational mobility. Focusing on the development of postgraduate education and continuing to promote the construction of world-class universities will help to alleviate the development dilemma caused by the current expansion of higher education. Moreover, the important role of family capital in people's access to higher education, especially quality higher education, should not be overlooked. Competition for quality higher education resources among capital-rich families may reinforce the social reproduction function of higher education and exacerbate intergenerational income inequality. Therefore, the future efforts of the Chinese government may focus on how to further promote equity in access to quality higher education by promoting its development.

There are several limitations to this study. While the methods used in this paper, including the Heckman sample selection model and the instrumental variables approach, reduce empirical estimation bias, they are unable to fully reduce the potential life-cycle problem and temporary income shocks. In addition, given the limited data, this paper does not identify a suitable proxy variable to replace individual ability and therefore cannot rule out the important influence of individual ability on intergenerational income mobility, which may lead to overestimation of the results of the empirical analyses.

CRediT authorship contribution statement

Nan Zhang: Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Hongmin Fan:** Writing – review & editing, Writing – original draft, Funding acquisition, Data curation, Conceptualization.

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Conflict of interest

The authors declare no conflict of interest.

References

- Alon, S., 2009. The evolution of class inequality in higher education: competition, exclusion, and adaptation. *Am. Sociol. Rev.* 74 (5), 731–755. <https://doi.org/10.1177/000312240907400503>.
- Ayalon, H., Yogeve, A., 2005. Field of study and students' stratification in an expanded system of higher education: the case of Israel. *Eur. Sociol. Rev.* 21 (3), 227–241. <https://doi.org/10.1093/esr/jci014>.
- Becker, G.S., Tomes, N., 1986. Human capital and the rise and fall of families. *J. Labor Econ.* 4 (3, Part 2), S1–S39. <https://doi.org/10.1086/298118>.
- Blanden, J., Machin, S., 2013. Educational inequality and the expansion of United Kingdom higher education. *Scott. J. Polit. Econ.* 60 (5), 597–598. <https://doi.org/10.1111/sjpe.12031>.
- Breen, R., Jonsson, J.O., 2007. Explaining change in social fluidity: educational equalization and educational expansion in twentieth-century Sweden. *Am. J. Sociol.* 112 (6), 1775–1810. <https://doi.org/10.1086/508790>.
- Breen, R., Luijkx, R., Müller, W., Pollak, R., 2009. Nonpersistent inequality in educational attainment: evidence from eight European countries. *Am. J. Sociol.* 114 (5), 1475–1521. <https://doi.org/10.1086/595951>.
- Chen, Y., Fan, Z., Gu, X., et al., 2020. Arrival of young talent: the send-down movement and rural education in China. *Am. Econ. Rev.* 110 (11), 3393–3430. <https://doi.org/10.1257/aer.20191414>.
- Chesters, J., Watson, L., 2013. Understanding the persistence of inequality in higher education: evidence from Australia. *J. Educ. Policy* 28 (2), 198–215. <https://doi.org/10.1080/02680939.2012.694481>.
- Chetty, R., Friedman, J.N., Rockoff, J.E., 2014. Measuring the impacts of teachers II: Teacher value-added and student outcomes in adulthood. *Am. Econ. Rev.* 104. <https://doi.org/10.1257/aer.104.9.2633>.
- Collins, R., 1977. Some comparative principles of educational stratification. *Harv. Educ. Rev.* 47 (1), 1–27. <https://doi.org/10.17763/haer.47.1.w55275200v4l4842>.
- Corak, M., 2013. Income inequality, equality of opportunity, and intergenerational mobility. *J. Econ. Perspect.* 27 (3), 79–102. <https://doi.org/10.1257/jep.27.3.79>.
- Dahl, M.W., DeLeire, T., 2008. The Association between Children's Earnings and Fathers' Lifetime Earnings: Estimates Using Administrative Data. *Inst Res Poverty, Univ Wisconsin-Madison*.
- Durkheim, E., 1956. *Education and Sociology*. Free Press, New York, NY.
- Emran, M.S., Shilpi, F.J., 2019. Economic approach to intergenerational mobility: Measures, methods, and challenges in developing countries (No. 2019/98). WIDER Working Paper. doi:10.35188/UNU-WIDER/2019/734-7.
- Fan, Y., Yi, J., Zhang, J., 2021. Rising intergenerational income persistence in China. *Am. Econ. J. Econ. Policy* 13 (1), 202–230. <https://doi.org/10.1257/pol.20170097>.
- Guo, Y., Song, Y., Chen, Q., 2019. Impacts of education policies on intergenerational education mobility in China. *China Econ. Rev.* 2019. <https://doi.org/10.1016/j.chieco.2019.03.011>.
- Guryan, J., Hurst, E., Kearney, M., 2008. Parental education and parental time with children. *J. Econ. Perspect.* 22 (3), 23–46. <https://doi.org/10.1257/jep.22.3.23>.
- Iannelli, C., 2011. Educational expansion and social mobility: the Scottish case. *Soc. Policy Soc.* 10 (2), 251–264. <https://doi.org/10.1017/S147474641000059X>.
- Iannelli, C., Smyth, E., 2008. Mapping gender and social background differences in education and youth transitions across Europe. *J. Youth Stud.* 11 (2), 213–232. <https://doi.org/10.1080/13676260701863421>.
- Jia, R., Li, H., Meng, L., 2022. Can Elite College Education Change One's Fate in China? Available at SSRN 4113768. <https://doi.org/10.2139/ssrn.4113768>.
- Lefgren, L., Sims, D., Lindquist, M.J., 2012. Rich dad, smart dad: decomposing the intergenerational transmission of income. *J. Polit. Econ.* 120 (2), 268–303. <https://doi.org/10.1086/666590>.
- Lindqvist, E., Vestman, R., 2011. The labor market returns to cognitive and noncognitive ability: evidence from the Swedish enlistment. *Am. Econ. J.: Appl. Econ.* 3 (1), 101–128. <https://doi.org/10.1257/app.3.1.101>.
- Lu, M., Zhang, X., 2019. Towards an intelligent country: China's higher education expansion and rural children's senior high school participation. *Econ. Syst.* 43 (2), 100694. <https://doi.org/10.1016/j.ecosys.2019.100694>.
- Lucas, S.R., 2001. Effectively maintained inequality: Education transitions, track mobility, and social background effects. *Am. J. Sociol.* 106 (6), 1642–1690. <https://doi.org/10.1086/321300>.
- Lv, G., Li, Y., 2017. Research on the variation of intergenerational income elasticity and its impact in China. *J. Xiamen Univ.: Philos. Soc. Sci.* (03), 35–45 (In Chinese).
- Ma, G., Ji, Y., Xu, J., 2017. How does university expansion affect the higher education premium? *Manag. World* 08, 60–71. <https://doi.org/10.19744/j.cnki.11-1235/f.2017.08.007> (In Chinese).
- Marginson, S., 2016. High participation systems of higher education. *J. High. Educ.* 87 (2), 243–271. <https://doi.org/10.1080/00221546.2016.11777401>.
- Marginson, S., 2018. Higher education, economic inequality and social mobility: Implications for emerging East Asia. *Int. J. Educ. Dev.* 2018, 4–11. <https://doi.org/10.1016/j.ijedudev.2017.03.002>.
- Mok, K.H., 2016. Massification of higher education, graduate employment and social mobility in the Greater China region. *Br. J. Sociol. Educ.* 37 (1), 51–71. <https://doi.org/10.1080/01425692.2015.1111751>.
- Müller, W., Pollak, R., 2004. Social mobility in West Germany: The long arms of history discovered. In: Breen, R. (Ed.), *Social Mobility in Europe*. Oxford University Press, New York, NY, pp. 77–113.
- Nunn, N., Qian, N., 2014. US food aid and civil conflict. *Am. Econ. Rev.* 104 (6), 1630–1666. <https://doi.org/10.1257/aer.104.6.1630>.
- Raftery, A.E., Hout, M., 1993. Maximally maintained inequality: expansion, reform, and opportunity in Irish education, 1921–75. *Sociol. Educ.* 66 (1), 41–62. <https://doi.org/10.2307/2112784>.
- Shavit, Y., Westerbeek, K., 1998. Reforms, expansion, and equality of opportunity. *Eur. Sociol. Rev.* 14 (1), 33–47. <https://doi.org/10.1093/oxfordjournals.esr.a018226>.
- Solon, G., 1992. Intergenerational income mobility in the United States. *Am. Econ. Rev.* 1992, 393–408. (<http://www.jstor.org/stable/2117312>).
- Stock, J.H., Yogo, M., 2002. Testing for weak instruments in linear IV regression. NBER Working Papers. DOI:10.3386/t0284.
- Yeung, W.J.J., 2013. Higher education expansion and social stratification in China. *Chin. Sociol. Rev.* 45 (4), 54–80. <https://doi.org/10.2753/CSA2162-0555450403>.
- Yuan, W., 2017. The Sins of the Fathers: intergenerational income mobility in China. *Rev. Income Wealth* 63 (2), 219–233. <https://doi.org/10.1111/roiw.12222>.
- Zhang, D., Li, L., 2019. A reanalysis of regional differences in enrollment expansion and access to higher education. *Peking Univ. Educ. Rev.* 17 (01), 142–162+191–192 (In Chinese).
- Zou, W., Ma, Z., 2019. Family background, intergenerational transmission and educational inequality. *China Ind. Econ.* 2019 (2), 80–98. <https://doi.org/10.19581/j.cnki.ciejournal.2019.02.012> (In Chinese).

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